

ONLINE - VIDEO BASED LEARNING PLATFORM

1. INNER JOIN

```
-- Check how many students has watch this video and what is the total watch
hour for the video
SELECT v.title AS 'Title',COUNT(*) AS 'Total Watch',
SEC_TO_TIME(SUM(TIME_TO_SEC(v.duration))) AS 'Total Duration' from
user_video_metrics as m INNER JOIN users AS u ON u.id = m.user_id INNER
JOIN videos AS v ON v.id = m.video_id WHERE m.video_id =1;
```

-- Output table

title	Total Watch	Total Duration
Intro to Python	1	00:45:00

-- Check for one user

```
SELECT u.id AS id, u.name AS Name ,COUNT(*) as 'Total Watch',
SEC_TO_TIME(SUM(TIME_TO_SEC(v.duration))) AS 'Total Duration' FROM
user_video_metrics as m INNER JOIN users as u ON u.id = m.user_id INNER
JOIN videos AS v ON v.id = m.video_id WHERE u.id = 1;
```

-- Output table

Name	Total Watch	Total Duration
Rinzin	3	00:25:60

2. INSERT STATEMENT

-- Insert new topic

```
INSERT INTO `topics` (`name`, `created_at`)
VALUES ('Database Design', NOW());
```

-- Insert new Video

```
INSERT INTO `videos` (`title`, `slug`, `topic_id`, `thumbnail`,
`video_url`, `duration`, `description`, `created_at`, `updated_at`)
VALUES (
  'Database Design',
  'database-design',
  51,
  'https://cdn.example.com/thumb/051.jpg',
  'https://cdn.example.com/video/051.mp4',
```

```
'01:00:00',  
'Build database schema.',  
NOW(),  
NOW()  
);
```

3. UPDATE STATEMENT

```
-- Update Topics  
UPDATE topics  
SET name = 'Database Design Course'  
WHERE id = 1;
```

```
-- Update videos  
UPDATE videos  
Set thumbnail = 'image'  
WHERE id = 1;
```

4. VIEW STATEMENT

```
-- View for the video and topics  
CREATE VIEW `video_with_topic` AS  
SELECT  
    v.id,  
    v.title,  
    v.slug,  
    t.name AS topic_name,  
    v.duration,  
    v.description,  
    v.thumbnail,  
    v.video_url,  
    v.created_at  
FROM `videos` v  
JOIN `topics` t ON v.topic_id = t.id;
```

```
-- We can select from this new view  
SELECT * FROM `video_with_topic` WHERE id = 2;
```

5. CUSTOM FUNCTION

```
DELIMITER $$
```

```
CREATE FUNCTION `get_video_stats`(p_video_id INT)
RETURNS VARCHAR(100)
DETERMINISTIC
BEGIN
    DECLARE v_count INT;
    DECLARE v_total TIME;

    SELECT COUNT(*) INTO v_count
    FROM user_video_metrics
    WHERE video_id = p_video_id;

    SELECT SEC_TO_TIME(SUM(TIME_TO_SEC(interact_At))) INTO v_total
    FROM user_video_metrics
    WHERE video_id = p_video_id;

    RETURN CONCAT('Total Interactions: ', v_count, ' | Total Watch
Duration: ', v_total);
END$$

DELIMITER ;

-- function
SELECT get_video_stats(1);
```